



Quantifying carbon emissions in cold chain transport: A real-world data-driven approach[☆]

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ABSTRACT

Sustainable urban development necessitates advanced low-carbon transportation strategies, particularly within the cold chain logistics sector, where transporting perishable goods significantly contributes to environmental emissions. This study addresses the gap in empirical research by deploying real-world data from 128 long-haul trips, provided by a logistics company, to develop a comprehensive bottom-up operational-level carbon emission model. Our model quantifies emissions through various operational phases—loading, unloading, and transport—capturing contributions from vehicle operation (driving and idling), refrigeration processes (including transmission, infiltration, and pre-cooling), and refrigerant leakage. It further assesses the impact of ambient temperature on emissions and examines the effectiveness of decarbonization strategies such as employing shore power for pre-cooling and adopting low-carbon refrigerants. Validated against actual fuel consumption with an impressive accuracy of −1.84%, our findings significantly advance green logistics practices, offering practical insights for a transition towards net-zero emissions and improving the sustainability of cold chain transportation systems.

1. Introduction

The 2015 Paris Agreement set forth an ambitious framework to combat climate change, aiming to maintain global temperature rise below 2 °C above pre-industrial levels and striving for a more stringent cap of 1.5 °C (UNFCCC, 2020). Across the globe, this goal has catalyzed researchers, policymakers, and governmental bodies to prioritize the reduction of greenhouse gas (GHG) emissions across various sectors. One notable contributor is the cold chain logistics sector, responsible for ensuring perishable goods remain within a designated temperature range throughout their transport and storage. Although essential, this sector is inherently energy-intensive, resulting in substantial GHG emissions (Bin et al., 2022). According to UNEP & FAO (2022), cold chain logistics are estimated to account for about 4 % of the global GHG emissions. Factors such as population growth, increased living standards, and a surge in online shopping due to post-pandemic shifts (Qin et al., 2019; Tsang et al., 2024) have spurred a consistent rise in demand for these services. The fleet of refrigerated vehicles is expected to grow significantly, from less than 3 million units in 2013 to an expected 15.5 million by 2025 (Fleet, 2015). Meanwhile, the market value of refrigerated transport is expected to grow at a compound annual growth rate

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(CAGR) of 7.2 %, reaching USD 160.7 billion by 2027 (Mehra, 2022). These trends underscore the urgent need for robust decarbonization strategies within this rapidly expanding sector to meet international climate goals.

Transport refrigeration units (TRUs), vital for controlling the temperature of goods during transit, are central to the operations within the cold chain. The carbon emissions from these vehicles predominantly stem from the energy used by both the vehicle engines and the TRU engines. Additionally, the refrigerants employed in these units—often hydrofluorocarbons with high global warming potential (GWP)—further exacerbate environmental impacts through leakage (Bin et al., 2022; Carlsson-Kanyama, 1998; Gloël et al., 2022; Tassou et al., 2009; Wu, Li, et al., 2022). Research led by Wu, Li, et al. (2022) introduced a carbon footprint model for refrigerated vehicles using Life Cycle Assessment (LCA), which revealed that the usage phase of these vehicles contributes to about 96 % of their total carbon emissions. This significant footprint is largely attributed to increased fuel consumption driven by the added weight and frontal area of the TRU, as well as the diesel engine's role in powering these units. Further compounding the emissions, Yang et al. (2021) found through the Passenger car and Heavy duty Emission Model (PHEM) that TRU-equipped vehicles emit 15 % more CO₂ and may increase NO_x emissions by up to 18 % compared to standard trucks. Other contributing factors include the temperature within the refrigerated compartment and ambient environmental conditions (Adekomaya et al., 2016; Rai, 2019; Rai & Tassou, 2017; Yang et al., 2021), potential refrigerant leaks (Gloël et al., 2022; Tassou et al., 2009; Wu et al., 2013), frequency of door openings (Rai, 2019; Yang et al., 2021), and variations in driver behavior (Demir et al., 2011; Yang et al., 2021). Notwithstanding the advancements in technology, the majority of refrigerated vehicles still depend on traditional fossil fuels (Rai, 2019; Rai & Tassou, 2017), further amplifying their environmental burden.

Existing studies on carbon emissions in cold chain logistics predominantly focus on comprehensive lifecycle assessments of perishable goods. For instance, Bin et al. (2022) applied an LCA to evaluate GHG emissions at various stages of the fruit and vegetable cold chain, revealing that the transportation phase is responsible for 82 % of total emissions. Shen et al. (2023) explored energy consumption patterns in an industrial cold food storage facility, highlighting that frozen foods necessitate substantially higher energy for refrigeration due to their lower target temperatures, with warehousing alone comprising nearly half of the total energy usage (47.81 %). Furthermore, du Plessis et al. (2023) assessed fuel consumption and related emissions in refrigerated road transport at the transport service level based on 147 long-distance trips, incorporating factors such as trailer type, route, load weight, and refrigeration duration, and achieved a prediction accuracy of 6.7 % for fuel consumption.

Research on carbon reduction in refrigerated transport explores two principal strategies: route optimization and refrigeration system improvements. The route optimization approach involves developing models that balance emissions with other logistics parameters (Chen et al., 2021; Liu et al., 2019; Qin et al., 2019; Zhang et al., 2023). For example, Qin et al. (2019) proposed a model optimizing cold chain vehicle routes by considering cost, customer satisfaction, and GHG emissions, aimed at minimizing the unit satisfaction cost. On the refrigeration side, alternative refrigerants and enhancements in system energy efficiency are investigated (Cardoso et al., 2017; Dong et al., 2021; Wu, Li, et al., 2022; Wu, Liu, et al., 2022; Wu et al., 2013). Wu et al. (2013) conducted an LCA on different refrigerants, suggesting R410A for high-temperature and R744 (CO₂) for low-temperature settings to optimize carbon reductions. Further, Wu, Liu, et al. (2022) compared traditional vapor compression refrigeration systems (VCRS) to phase change material-based cold storage systems (PCCSS), finding a reduction in carbon emissions by 22 % to 56 % with PCCSS, primarily during the usage phase, influenced by factors such as engine efficiency, energy mix, and the GWP of refrigerants.

While the widespread adoption of electrified refrigerated trucks is still in its nascent stages, research into the carbon reduction potential of this transition is growing. Thornton et al. (2018) investigated the use of electric Transport Refrigeration Units (eTRUs) in the United States, noting that electric refrigerated trucks, when connected to the electrical grid, could supplant diesel engines during idling periods, resulting in substantial carbon emission reductions. Similarly, Zhang et al. (2023) assessed the combined use of electric and diesel refrigerated trucks within urban cold chain distribution networks. Utilizing a multi-objective optimization model, their study illustrates how government subsidies and carbon emission trading schemes can optimize distribution routes and vehicle choices, significantly lowering both carbon emissions and operational costs.

Despite substantial research into the environmental impact of refrigerated transport and various carbon reduction strategies, there remains a notable gap in operational-level evaluation, dissecting carbon emissions across distinct operational stages of cold chain transportation. This oversight primarily stems from the scarcity of granular operational data. This detailed analysis is crucial for identifying specific emission reduction opportunities and enhancing sustainability strategies within logistics. However, recent advancements in Internet of Things (IoT) technology, which facilitate the collection of real-time and real-world data, are paving the way for transformative improvements in green logistics. The adoption of IoT-driven, data-centric approaches is anticipated to significantly enhance operational efficiencies and sustainability within logistics systems (Liu et al., 2019).

This study capitalizes on advancements in IoT technology by engaging in a comprehensive case study with a logistics company in Taiwan. This research is distinct as it quantifies carbon emissions across each operational phase of cold-chain logistics transport using real-time IoT data, offering a level of detail not typically achieved in the existing literature. We develop a bottom-up carbon emission model specifically designed to identify emission hotspots and assess the impacts of driving, idling, refrigeration, and refrigerant leakage with unprecedented granularity. This approach facilitates the identification of key areas for emission reduction, significantly advancing the discourse on logistical carbon footprint management. Moreover, the study evaluates targeted decarbonization strategies—such as the adoption of low-carbon refrigerants and the use of shore power for pre-cooling. These strategies are developed in direct collaboration with the logistics provider to ensure they are not only theoretically valid but also practically feasible and immediately implementable. By bridging a critical research gap with detailed operational-level analysis, this work enhances understanding of carbon emissions in cold-chain logistics. Our comprehensive examination not only aligns with global climate mitigation efforts but also sets a practical benchmark for industry adoption, thus significantly enhancing the literature on sustainable logistics practices.

2. Methodology and data

2.1. Data collection and visualization

To understand the logistics operations of the case company in Taiwan, which specializes in the delivery of fresh goods to convenience stores, this study analyzes the use of two types of diesel-powered refrigerated trucks with different capacities: a 3.32-ton truck (Truck3) and a 5.75-ton truck (Truck5). Both trucks are equipped with vapor compression TRUs utilizing R404A refrigerant. The refrigeration systems are powered by a single engine that also drives the trucks, integrating propulsion and refrigeration functions. These vehicles also feature advanced tracking technologies, such as second-by-second GPS tracking, cargo temperature sensors, and geofencing capabilities, enabling the collection of detailed operational data including fuel tank levels to verify fuel consumption in our carbon emission models.

Our analysis draws on data from 128 long-haul deliveries conducted between December 2023 and April 2024, during which ambient temperatures ranged from 11 °C to 29 °C. The deliveries using Truck3 and Truck5 are differentiated by vehicle type and storage temperature conditions—cool storage at 18 °C and cold storage at 3 °C for Truck3, and cool storage at 18 °C for Truck5. These truck types are chosen due to their widespread use in small-scale freight transport within Taiwan's cold chain logistics sector, representing common fleet configurations. The selected storage temperatures of 18 °C for cool and 3 °C for cold storage align with industry-standard practices, effectively reflecting the typical operational scenarios in this sector. This robust carbon emission modeling initiative capitalizes on real-world delivery data, which is comprehensively cataloged in [Table 1](#). This table not only details vehicle characteristics but also provides descriptive statistics for each trip, such as average delivery distance, duration, and the number of delivery locations. Incorporating these precise, real-world data points ensures that our findings offer practical, universally applicable insights that are directly relevant to optimizing cold chain logistics operations.

[Fig. 1](#) presents a sample of the Truck3 (18 °C) telematics data, illustrating the relationship between speed, temperature, fuel level, and time. This detailed data helps in understanding the operational status of the vehicles. For instance, the starting and refueling of the vehicle around 6 a.m. initiate the pre-cooling process. The internal temperature is then moderated from the previous day's residual heat to the target temperature for loading. Fluctuations in the cargo compartment temperature occur due to external disturbances like door openings.

[Fig. 2](#) illustrates a typical logistics route for cold chain operations within northern Taiwan, utilized by the case study company. The map highlights the starting depot, various delivery points, and the routes trucks follow throughout the region. This diagram enhances understanding of the logistical complexities faced in distributing perishable goods. The top panel of [Fig. 2](#) outlines the sequence of cold chain delivery operations, beginning at the depot where trucks are loaded and the TRU is activated. Following this, trucks visit multiple delivery sites (indicated as N stops), where goods are offloaded at each designated location. After completing deliveries, the TRU is deactivated, and the trucks return empty to the depot, concluding the cycle.

Following the analysis of the three delivery types and their respective stages, we delve into the allocation of time across the loading, unloading, and transport phases, as shown in [Fig. 3](#). Truck5 (18 °C) notably spends the most time on deliveries due to longer travel routes and more frequent stops. Further scrutiny reveals that a substantial portion of the delivery duration—between 60.0 % and 74.0 %—is spent idling. This idle time primarily occurs during extended loading and unloading periods at warehouses and retail outlets, highlighting areas where efficiency improvements could significantly reduce overall delivery times.

2.2. Bottom-up carbon emission model

This subsection introduces the bottom-up operational-level carbon emission model developed to specifically quantify emissions from transportation activities within cold chain logistics. The model integrates data across various operational stages, enabling a detailed evaluation of carbon emissions for each segment involved in the logistics process. [Fig. 4](#) illustrates the framework of the

Table 1
Specifications and delivery data for vehicles used in this study.

Parameter Group	Detail	Vehicle Types of Delivery		
		Truck3		Truck5
		Cool (18 °C)	Cold (3 °C)	Cool (18 °C)
Vehicle Characteristics	Vehicle class	Euro 6 diesel truck		Euro 5 diesel truck
	Vehicle mass (kg)	3320		5750
	Loading capacity (kg)	1680		3750
	Refrigerant weight (R404A) (kg)	1.4		2.6
Delivery Data Information	Delivery temperature (°C)	18		3
	Number of samples	75	13	40
	Average distance (km)	78.11	70.81	87.75
	Average duration (hour)	4.86	3.97	7.36
	Average number of stops	23	10	39

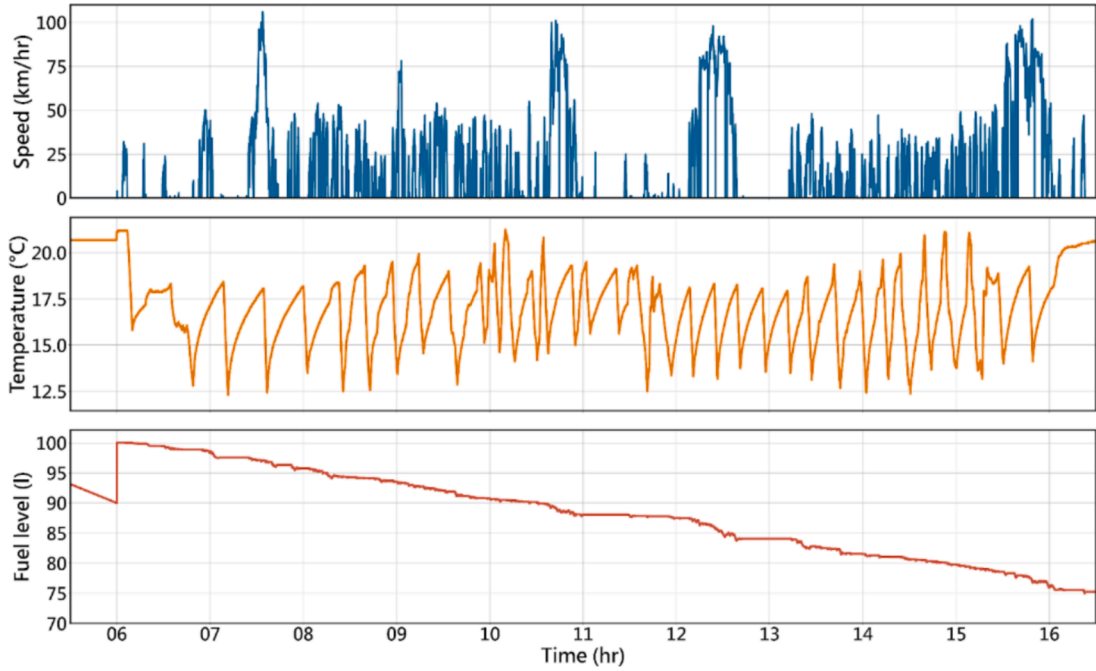


Fig. 1. Time series plots of speed, cargo temperature, and fuel consumption for Truck 3 (18 °C) during a delivery sample collected on April 27, 2024.

carbon emission model developed in this study. Model inputs consist of vehicle GPS and telematics data, capturing details like speed, fuel level, and geolocation. Additional data, including vehicle weights and refrigerant types, are sourced from vehicle catalogs, complemented by environmental data from the Central Weather Bureau (CWB). These diverse inputs feed into a calculation module divided into three primary segments: fuel consumption during vehicle operation, fuel consumption related to refrigeration, and emissions from refrigerant leakage. Each segment is elaborated upon in subsequent sections (2.2.1 to 2.2.3).

The parameters involved in the emissions calculations are detailed in Table 2, which includes descriptions, values, and sources covering aspects related to vehicle operation, refrigeration, and refrigerant leakage. This table serves as a foundational dataset for the model's computations. Model validation is conducted by comparing the predicted fuel consumption with actual sensor data from the vehicles, detailed further in Section 2.3. The final output of the model provides a detailed breakdown of carbon emissions across three different delivery vehicle types, each segmented by operational phase. This granular analysis underscores significant opportunities for emission reductions within logistics operations, guiding targeted strategies to minimize environmental impact.

2.2.1. Vehicle operation (driving and idling)

Carbon emissions from vehicle operations, encompassing driving and idling, are a significant contributor to the overall emissions profile of cold chain logistics. Idling, where the engine runs without the vehicle moving, historically accounted for approximately 9 % of total transport emissions (Bhandari et al., 2013). This underscores the necessity of using precise, microscopic emission models for accurate assessments.

For this study, the Comprehensive Modal Emissions Model (CMEM), developed by Barth et al. (2005), is chosen due to its flexibility in adapting to various engine parameters and its efficacy across different vehicle types, from heavy-duty diesel vehicles to light-duty trucks (Talley et al., 2016). CMEM calculates the power required for driving ($P_{driving}$) by factoring in the vehicle's speed, weight, and the slope of the road as outlined in Equation (1). The payload of the vehicle parameter is derived from company-provided data on the initial total weight of goods for each trip, which is assumed to be evenly distributed across all delivery locations to estimate the changing payload throughout the operations. Furthermore, the model considers the power demands from auxiliary vehicle systems such as air conditioning to be negligible, aligning with findings from Demir et al. (2011).

Equation (2) then computes the instantaneous fuel usage rate ($F_{driving}$) based on the derived power demand, accurately capturing the dynamics of fuel consumption during active vehicle operation and idling phases. When the vehicle remains stationary, the model attributes all fuel consumption to idling, ensuring that emissions calculations reflect real-world operating conditions. Finally, Equation (3) converts the calculated fuel consumption into carbon dioxide equivalent emissions ($Emission_{driving}$), providing a quantifiable measure of the vehicle's environmental impact during its operational lifecycle. This conversion uses established emission factor (EF_{diesel}), facilitating comparisons and benchmarking against regulatory standards and previous studies. Detailed parameters utilized in the model calculations are listed in Table 2.

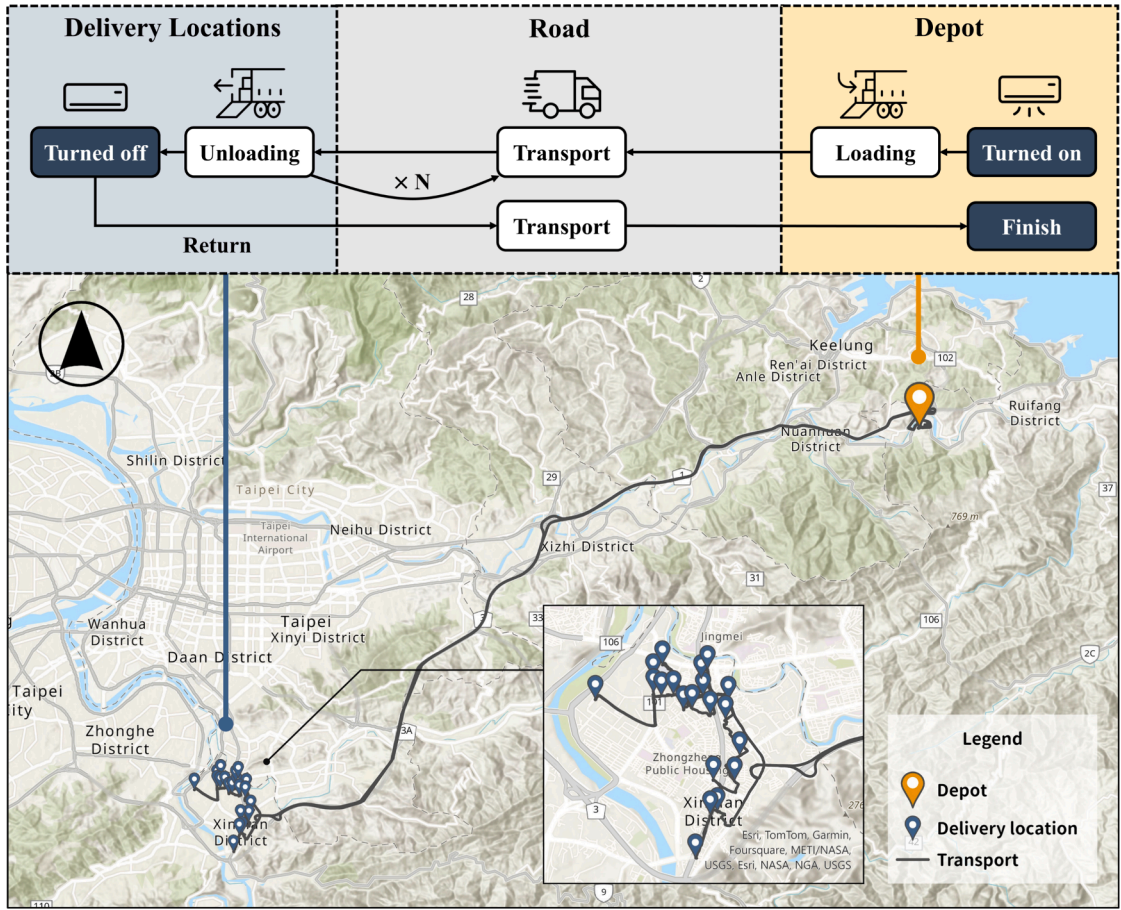


Fig. 2. Route and operational workflow of typical cold chain transportation for a logistics company in Taiwan, highlighting the depot, multiple delivery sites, and the network of transport paths.

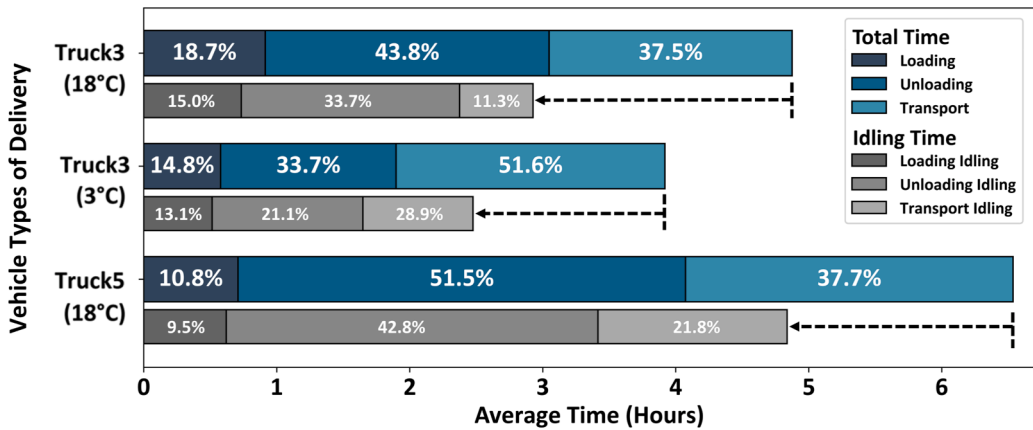
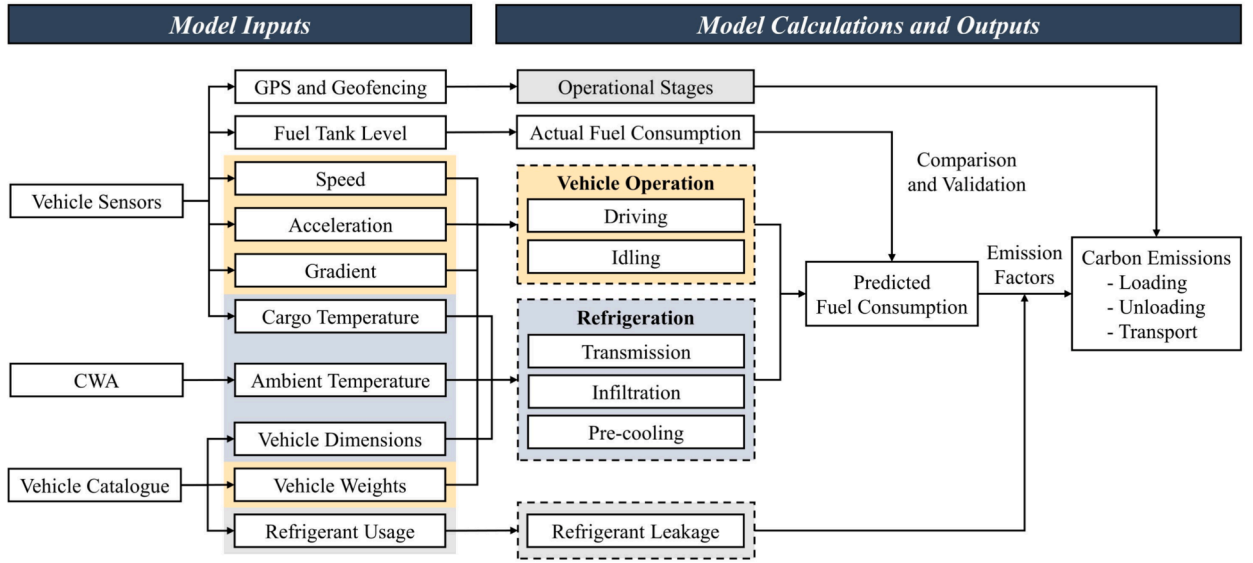


Fig. 3. Distribution of average time spent on various operational phases for three vehicle types of cold chain deliveries, detailing periods of activity and idling.

$$P_{driving} = \frac{0.5C_d\rho_a A_f v^3 + (W_v + W_f)v(a + g\sin\theta + gC_r\cos\theta)}{1000\epsilon} \quad (1)$$



* CWA : Central Weather Administration

Fig. 4. Framework of the bottom-up carbon emission model, showing the integration of inputs like sensor data and environmental conditions, calculation modules by delivery segment, and the resulting emission outputs across operational phases.

$$F_{driving} = \frac{\xi}{\kappa\psi} \left(kNV + \frac{P_{driving}}{\varpi} \right) \quad (2)$$

$$Emission_{driving} = F_{driving} \times EF_{diesel} \quad (3)$$

where $P_{driving}$ is the engine power demand (kW); C_d is the coefficient of aerodynamic drag; ρ_a is the air density (kg/m^3); A_f is the frontal surface area (m^2); v is the speed (m/s); W_v and W_l are the vehicle gross and payload weights (kg); a is the acceleration (m/s^2); g is the gravitational constant (m/s^2); θ is the road gradient (degrees); C_r is the coefficient of rolling resistance; ϵ is the vehicle drive train efficiency (—); $F_{driving}$ is the fuel usage rate (liter/s); ξ is the fuel-to-air mass ratio (—); κ is the heating value for diesel fuel (kJ/g); ψ is the conversion factor from grams to liters (liter/g); k is the engine friction factor ($\text{kJ}/\text{liter} \cdot \text{s}$); N is the engine speed (rev/s); V is the engine displacement (liters); ϖ is the efficiency parameter for diesel engines (—); $Emission_{driving}$ is the carbon emissions during driving (kg CO₂e); EF_{diesel} is the carbon emission conversion factor for diesel (kg CO₂e/liter).

2.2.2. Refrigeration (transmission, infiltration, and pre-cooling)

In refrigerated transport, the TRU is essential for maintaining optimal cargo space temperatures by mitigating thermal loads. This study utilizes a refrigeration model based on guidelines from the American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) along with recent findings by Bin et al. (2022). The model calculates the thermal load (Q_{ref}) by considering various sources of heat (Equation (4)): transmission (Q_{trans}), which involves heat entering through the truck's surfaces; infiltration (Q_{inf}), which accounts for heat introduced through air exchange during door openings; and pre-cooling (Q_{pre}), which removes heat from the insulated box and its internal air. Unlike typical scenarios involving bulk transport of homogeneous products, such as fruits and vegetables (Rai, 2019; Rai & Tassou, 2017; Yang et al., 2021), this study excludes product load due to the nature of the goods—diverse, pre-packaged items already conditioned in a cold storage facility, which significantly reduces additional cooling requirements. Equations (5) through (8) detail the calculations for these thermal components.

$$Q_{ref} = Q_{trans} + Q_{inf} + Q_{pre} \quad (4)$$

where

$$Q_{trans} = \sum U \cdot A \cdot (T_{amb} - T_{int}) \quad (5)$$

$$U = \frac{1}{\sum_{i=1}^n \frac{x_i}{k_i}} \quad (6)$$

$$Q_{inf} = T \cdot 0.577 \cdot W \cdot H^{1.5} \cdot \left(\frac{Q_s}{A} \right) \cdot \left(\frac{1}{R_s} \right) \quad (7)$$

Table 2

Description of parameters used in the model.

Parameter	Description	Value	Source
Vehicle Operation (Driving and Idling)			
W_v	Gross vehicle weight (kg)	3320 (Truck3) 5750 (Truck5)	Company data
W_l	Payload of vehicle (kg)	NA ^a	Company data
A_f	Frontal surface area (m ²)	3.762 (Truck3) 5.405 (Truck5)	Company data
V	Engine displacement (liters)	5.193	Company data
v	Speed (m/s)	NA ^a	Sensors
a	Acceleration (m/s ²)	NA ^a	Sensors
θ	Road gradient (degrees)	NA ^a	Computed from GPS
N	Engine speed (rev/s)	NA ^a	Sensors
ξ	Fuel-to-air mass ratio (–)	1	Franceschetti et al. (2013)
κ	Heating value for diesel fuel (kJ/g)	44	
ψ	Conversion factor from grams to liters (liter/g)	737	DESNZ (2023)
k	Engine friction factor (kJ/liter ·s)	0.2	
ρ_a	Air density (kg/m ³)	1.2041	
C_d	Coefficient of aerodynamic drag	0.7	
C_r	Coefficient of rolling resistance	0.01	
g	Gravitational constant (m/s ²)	9.81	
ϵ	Vehicle drive train efficiency (–)	0.4	
σ	Efficiency parameter for diesel engines (–)	0.9	
EF_{diesel}	Diesel emission factor (kg CO2e/liter)	2.66	
Refrigeration (Transmission, Infiltration, and Pre-cooling)			
U	Overall heat transfer coefficient (W/(m2 ·K))	0.33	Company data
A	Area of surface (m ²)	41.050 (Truck3) 56.845 (Truck5)	Company data
T_{amb}	Ambient/external temperature (K)	NA ^a	Central Weather Administration, MOTC
T_{int}	Represents the internal temperature (K)	NA ^a	Sensors
x_i	Thickness of material (m)	0.1	Company data
k_i	Thermal conductivity of the materials (W/(m ·K))	0.033	Company data
V_{load}	Payload volume (m3)	12.27 (Truck3) 21.20 (Truck5)	Company data
C_{pa}	Specific heat of air (kJ·kg ^{−1} ·K ^{−1})	1.005	McMurry (2024)
T_{ini}	Initial temperature inside the body (K)	NA ^a	Sensors
T_{set}	Thermostat setting temperature (K)	NA ^a	Sensors
T	Door opening time (s)	NA ^a	Computed from sensor
W	Doorway width (m)	0.8 (Truck3) 1.6 (Truck5)	Company data
H	Doorway height (m)	1.5 (Truck3) 1.8 (Truck5)	Company data
$\frac{Q_s}{A}$	Sensible heat load of infiltration air per square meter of doorway opening (kW·m ^{−2})	4 (18 °C) 10 (3 °C)	ASHRAE (2018)
R_s	Sensible heat ratio of the infiltration air heat gain	0.48	ASHRAE (2018)
q	Diesel calorific value (kJ/l)	35145.6	Energy Administration, MOEA
η	Efficiency of diesel-engine generator	0.4	Bin et al. (2022)
COP_{diesel}	Coefficient of performance of auxiliary diesel engine refrigeration	2.071 (Truck3) 2.056 (Truck5)	Company data
$Capacity_{plug}$	The cooling capacity of plug-in refrigeration (kWh /kWh of grid power)	2.862	Company data
$EF_{electricity}$	Electricity emission factor (kg CO2e/kWh)	0.495	Energy Administration (2024)
Refrigerant leakage			
R	Refrigerant charge (kg)	1.4 (Truck3) 2.3 (Truck5)	Company data
α	Annual leakage rate (%)	10	Rai and Tassou (2017)
L	Annual operating time (s)	12 months x 22 days x 8 h x 3600 s	Inferred from working hours

^a The parameter is not a single value.

$$Q_{pre} = \frac{(V_{load} \cdot \rho_a \cdot C_{pa}) \cdot (T_{ini} - T_{set})}{1000} \quad (8)$$

These thermal loads are then analyzed for their implications on fuel consumption (F_{ref}) and corresponding carbon emissions ($Emission_{ref}$), detailed in Equations (9) and (10).

$$F_{ref} = \frac{Q_{ref}}{q \cdot \eta \cdot COP_{diesel}} \quad (9)$$

$$Emission_{ref} = F_{ref} \times EF_{diesel} \quad (10)$$

Additionally, the model explores the benefits of using plug-in refrigeration systems during the pre-cooling phase, translating thermal loads into electricity demands (P_{ref}) and related carbon emissions ($Ele_Emission_{ref}$) with Equations (11) and (12). Detailed parameter descriptions used in these calculations are provided in Table 2, ensuring a comprehensive understanding of how each factor contributes to the overall thermal dynamics of refrigerated transport operations.

$$P_{ref} = \frac{Q_{ref}}{Capacity_{plug}} \quad (11)$$

$$Ele_Emission_{ref} = P_{ref} \times EF_{electricity} \quad (12)$$

where Q_{ref} is the total thermal load (kWh); Q_{trans} is the transmission load (kWh); Q_{pre} is the precooling load (kWh); Q_{inf} is the infiltration load (kWh); U is the overall heat transfer coefficient ($W/(m^2 \cdot K)$); A is the area of surface (m^2); T_{amb} and T_{int} are the ambient and internal temperature, respectively (K); x_i is the thickness of insulated box (m); k_i is the thermal conductivity of materials ($W/(m \cdot K)$); T is the duration of door openings (s); W and H are the width and height of the doorway (m); $\frac{Q_s}{A}$ is the sensible heat load from infiltration air per square meter of doorway ($kW \cdot m^{-2}$); R_s is the sensible heat ratio, indicating the proportion of total heat gain that is due to sensible heat (-); V_{load} is the payload volume (m^3); ρ_a is the air density (kg/m^3); C_{pa} is the specific heat of air ($kJ \cdot kg^{-1} \cdot K^{-1}$); T_{ini} and T_{set} are the initial and set temperature inside the refrigeration unit, respectively (K); F_{ref} is the fuel consumption for refrigeration (liter); q is the diesel heating value (kJ/l); η is the efficiency of the diesel-engine generator (-); COP_{diesel} is the coefficient of performance of the diesel-powered refrigeration system, illustrating its efficiency (-); $Emission_{ref}$ is the carbon emissions from refrigeration (kg CO₂e); P_{ref} is the electricity demand for plug-in refrigeration systems during the pre-cooling phase (kW); $Capacity_{plug}$ measures the cooling efficiency of plug-in systems, reflecting the amount of cooling provided per kWh of electrical power consumed (kWh /kWh of grid power); $Ele_Emission_{ref}$ is the carbon emissions from electricity usage for refrigeration (kg CO₂e); $EF_{electricity}$ is the carbon emission conversion factor for electricity (kg CO₂e/kWh).

2.2.3. Refrigerant leakage

Refrigerant leakage represents a significant source of greenhouse gases, contributing substantially to the global warming potential of refrigeration systems (March Consulting Group, 1998). TRUs are particularly susceptible to higher emission rates than stationary systems due to continuous vibrations from road travel and exposure to varying weather conditions and temperatures (Tassou et al., 2009). Research by Gloël et al. (2022) showed that even new refrigeration trucks may have initial leakage rates as high as 13 %, a rate that typically rises as the equipment ages. By the end of their operational lifespan, these systems are likely to release any remaining refrigerants, exacerbating the issue further.

This study adopts the methodology outlined by Wu et al. (2013) for calculating carbon emissions from refrigerant leakage. The formula used, as shown in Equation (13), determines the emissions ($Emission_{leakage}$) based on the refrigerant type, the quantity of leakage, and the duration of TRU operation. The refrigerant R404A, frequently used in refrigerated trucks (Bin et al., 2022) and noted for its high global WGP of 3922 (Rai & Tassou, 2017), is the focus of this analysis. The detailed parameters and settings employed in these calculations are listed in Table 2.

$$Emission_{leakage} = \frac{(R \times \alpha \times GWP)}{L} \quad (13)$$

where $Emission_{leakage}$ is the carbon emissions from refrigerant leakage (kg CO₂e); R is the refrigerant charge (kg); α is the annual leakage rate (%); L is the annual operating time (s).

2.3. Model performance evaluation

To ensure the robustness and reliability of our carbon emission model, its performance is evaluated by comparing predicted against actual fuel consumption data gathered via vehicle sensors. The accuracy of the model is assessed through three key metrics: the percentage gap (Gap), absolute percentage error (APE), and mean squared error (MSE), each defined in Equations (14)–(16). These metrics are employed collectively to provide a thorough evaluation of the model's performance, ensuring its validity across different operational scenarios.

$$Gap(\%) = 100 \times \frac{(\hat{y}_i - y_i)}{y_i} \quad (14)$$

$$APE(\%) = 100 \times \frac{|\hat{y}_i - y_i|}{y_i} \quad (15)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (16)$$

where $\hat{y}_i$ is the predicted fuel consumption by the model; y_i is the actual fuel consumption measured by sensors, n denotes the total number of observations.

3. Results and discussion

3.1. Carbon emission breakdown analysis

Fig. 5(a) illustrates the average carbon emissions from various sources across three different delivery vehicle types, with Truck5 (18 °C) recording the highest total emissions. Despite having the same target temperature as Truck3 (18 °C), Truck5 (18 °C) possesses a larger compartment volume and surface area, significantly increasing transmission and infiltration emissions. As a result, the total refrigeration-related emissions of Truck5 (18 °C) are approximately 3.16 times those of Truck3 (18 °C). Moreover, due to its heavier mass, Truck5 (18 °C) generates 2.4 times the driving-related emissions compared to Truck3 (18 °C). In contrast, Truck3 (3 °C) displays lower overall emissions than Truck3 (18 °C) because it covers shorter average driving distances (referenced in Table 1). However, due to the lower set temperature, refrigeration emissions from Truck3 (3 °C) are around 1.13 times higher than those from Truck3 (18 °C).

In terms of emission source contribution, driving remains the predominant factor, constituting between 62.8 % and 66.9 % of total emissions. Although idling uses less fuel, the extensive idling time during deliveries contributes significantly to emissions, ranging from 9.0 % to 13.9 %. This highlights the importance of optimizing routes to reduce both driving and idling times, thereby lowering emissions. Additionally, reducing idling periods is a key strategy for cutting emissions in logistics operations. Refrigeration processes contribute approximately 18.6 % to 24.5 % of total emissions, with transmission through the vehicle's surfaces being the primary source, accounting for 12.9 % to 14.1 %. Infiltration emissions vary based on door opening frequency and set interior temperatures, with higher rates for trucks maintaining lower temperatures; Truck3 (3 °C) experiences an infiltration rate of 7.5 %, compared to 4.5 % for Truck3 (18 °C). Despite the smaller temperature differential inside Truck5 (18 °C), its longer door opening times due to more frequent stops result in infiltration rates ranging from 4.5 % to 10.5 %. Pre-cooling emissions, directly linked to the target temperature setting of the trucks, range from 0.9 % to 1.5 %. Lastly, refrigerant leakage, tied to the cooling operation duration, is highest in Truck3 (3 °C) at 2.3 % due to the colder set temperature, while the rates in other trucks are below 0.6 %.

Panel (b) of Fig. 5 presents emissions per unit product weight, highlighting significant differences in carbon efficiency across the vehicles. Truck5 (18 °C) shows a 27.91 % reduction in emissions per unit product weight compared to Truck3 (18 °C). This enhanced efficiency can be primarily attributed to Truck5's larger cargo capacity, which facilitates more effective emissions performance relative to the weight of the product transported. Conversely, while Truck3 (3 °C) exhibits lower overall emissions than Truck3 (18 °C), its

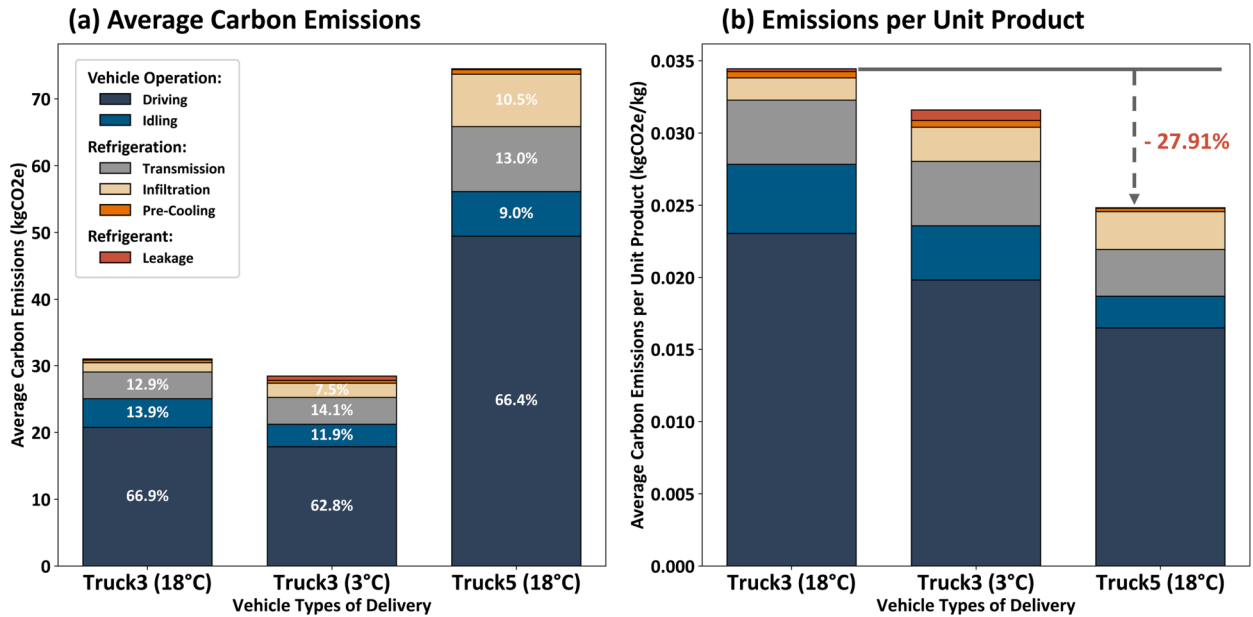


Fig. 5. (a) Average carbon emissions (kgCO₂e) per delivery and (b) emissions per unit product weight (kgCO₂e/kg) across different delivery segments for three types of cold chain transportation vehicles.

emissions per unit product weight are higher due to the additional energy needed to maintain a lower target temperature of 3 °C. These findings emphasize the benefits of larger vehicles and the importance of strategic vehicle selection and temperature management in reducing emissions per unit product in cold chain logistics.

Further analysis of the distribution and sources of carbon emissions across different operational phases enables the identification of specific stages where targeted emission reduction strategies can be effectively implemented. Fig. 6 illustrates that the transport phase is the largest contributor to emissions, accounting for between 72.3 % and 74.3 % of the total. Following this, the unloading phase contributes 19.6 % to 22.5 % of emissions, while the loading phase has the least impact, with emissions ranging from 5.2 % to 7.1 %. This underscores the significant impact of vehicle movement on fuel consumption during the transport phase, which, despite its shorter duration, results in higher emissions due to the intensive fuel consumption required.

In terms of contributors to emissions, idling predominates during the loading and unloading phases, reflecting prolonged periods where trucks remain stationary. Conversely, the transport phase is driven largely by emissions from driving, which constitute 85.1 % to 91.8 % of this phase's emissions. Infiltration significantly affects the unloading phase due to frequent door openings, particularly in larger trucks or those required to maintain colder operational temperatures. Conversely, no infiltration occurs during loading since trucks are loaded directly from a cold warehouse.

The carbon emissions during the transport phase are predominantly influenced by delivery mileage. A closer examination of emissions per kilometer reveals that Truck3, operating at both 18 °C and 3 °C, emits approximately 0.30 kgCO₂e/km. In comparison, Truck5 (18 °C) produces nearly double that amount, with emissions reaching 0.61 kgCO₂e/km, or 1.03 times more than Truck3. This substantial disparity highlights the importance of developing emission reduction strategies that account for vehicle size and operational conditions, thereby ensuring more targeted and effective measures to minimize carbon footprints.

3.2. Impact of ambient temperature on carbon emissions

This section explores how variations in ambient temperature influence carbon emissions from refrigeration and refrigerant leakage in refrigerated truck deliveries. The analysis primarily focuses on Truck3 (18 °C) and Truck5 (18 °C) due to the availability of a comprehensive data set for these specific configurations.

Fig. 7 details the correlation between ambient temperature and carbon emissions for both truck models. For Truck3 (18 °C), depicted in Fig. 7(a), an increase in ambient temperature correlates with rising emissions from both refrigeration and refrigerant leakage. The data illustrates a potential increase in total refrigeration emissions by up to 4.5 times, with transmission emissions particularly sensitive to changes in ambient temperature. At the highest measured ambient temperatures, transmission emissions reach their peak, decreasing sharply as temperatures fall below 15 °C. When ambient temperatures are between 11 °C and 13 °C, emissions drop to near-negligible levels, reflecting the minimal cooling required. Refrigerant leakage emissions increase with ambient

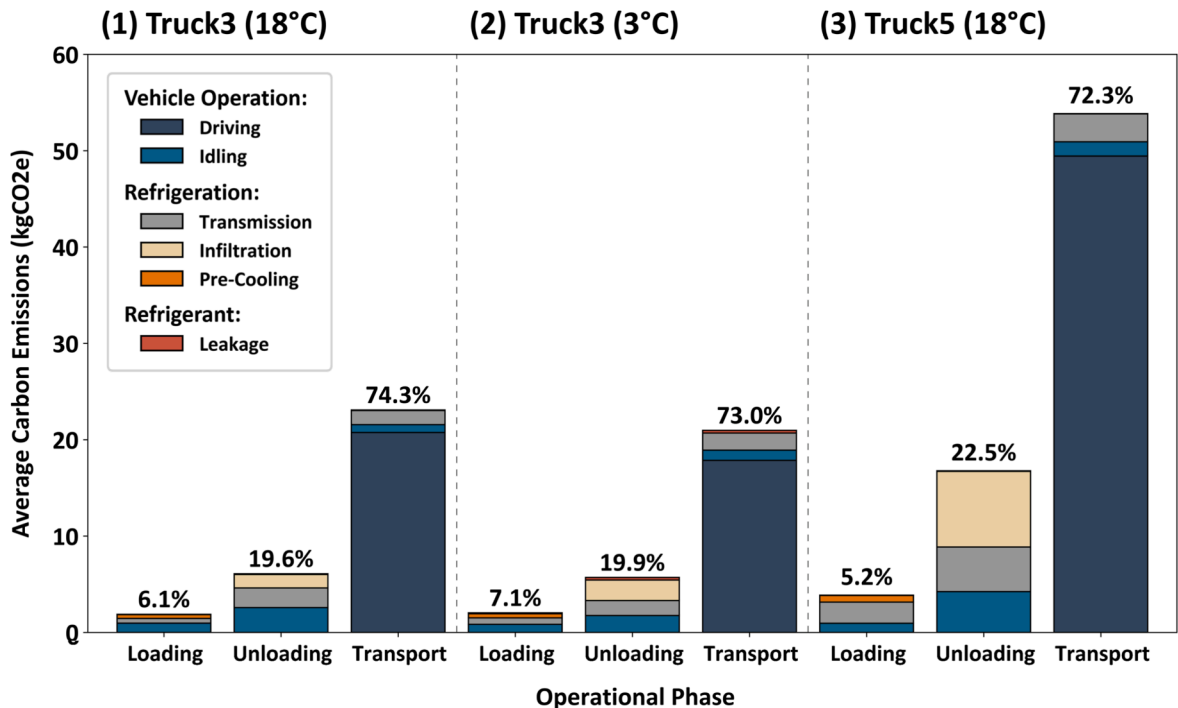


Fig. 6. Average carbon emissions by operational phase for three vehicle types, displaying the distribution across different segments of the delivery process.

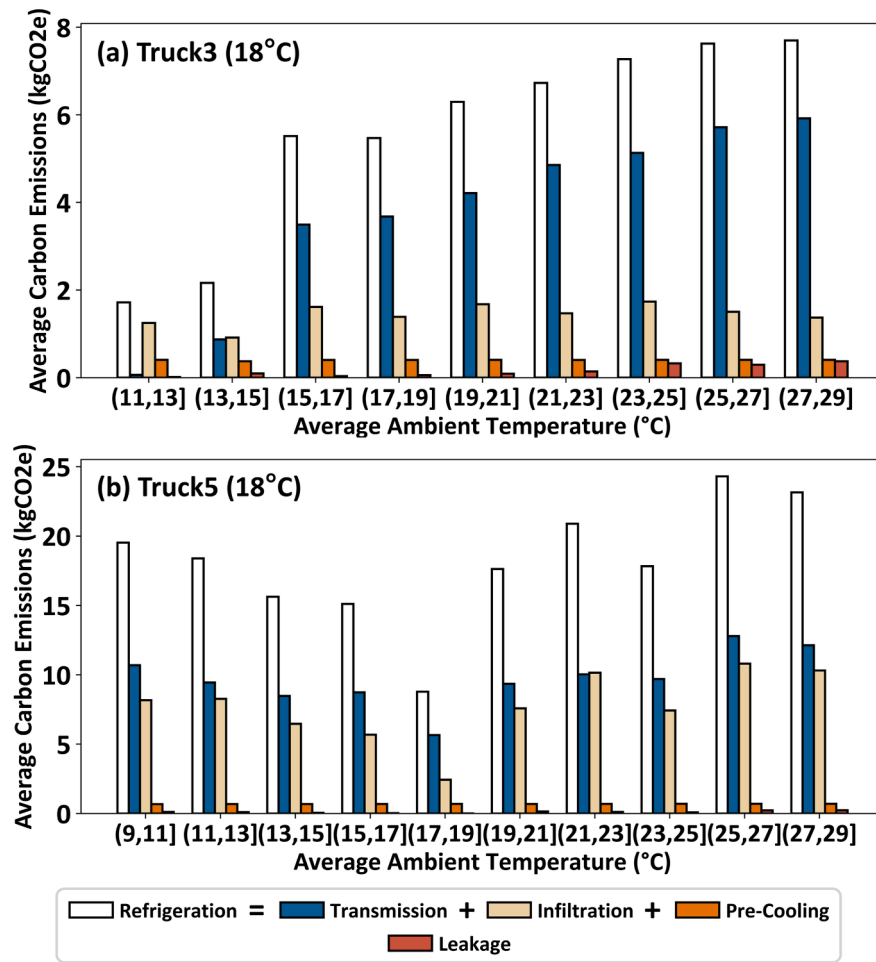


Fig. 7. Variability in carbon emissions across different ambient temperatures for refrigeration and refrigerant leakage: (a) Truck3 (18 °C) and (b) Truck5 (18 °C).

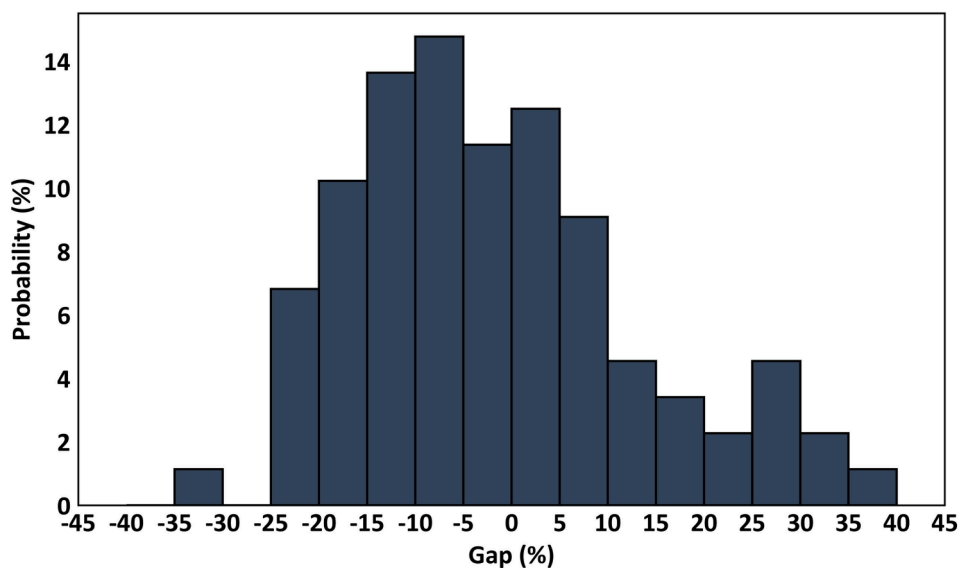


Fig. 8. Distribution of model accuracy gap (%).

temperature due to extended cooling periods necessitated by higher external temperatures.

In contrast, Fig. 7(b) shows the emissions profile for Truck5 (18 °C), which often operates during cooler night temperatures. In these conditions, the TRU's reverse-cycle mode activates to heat the cargo area to the set temperature of 18 °C, leading to increased transmission and refrigerant leakage emissions as the internal temperature aligns with the colder external environment. Similar to Truck3, infiltration emissions are less influenced by ambient temperatures and more by the frequency and duration of door openings, which emphasizes the role of operational practices in emissions. Pre-cooling emissions remain largely unaffected by changes in ambient temperature due to stable initial and target temperatures.

3.3. Model performance in fuel consumption prediction

Given the absence of a fuel consumption sensor in Truck5, our evaluation focuses on data collected from 88 trips using Truck3. Fig. 8 displays the variation in the percentage gap between the model's predictions and actual fuel consumption, ranging from −35 % to +40 %. Despite this variation, the model consistently demonstrates high accuracy, with an average gap of only −1.84 %. This precise prediction ability, verified by an APE of 12.25 %, is pivotal for managing carbon emissions effectively in cold chain logistics. Furthermore, the incorporation of the MSE of 3.43 enriches the robustness of our evaluation, providing a more comprehensive understanding of the model's performance across varied operational scenarios. These metrics collectively affirm the model's precision in forecasting fuel consumption.

3.4. Sensitivity of carbon emissions to key operational parameters

Our model, primarily built on real-world parameters, incorporates sensitivity adjustments to rigorously evaluate potential variations. For the sensitivity analysis, we select the delivery sample with the highest precision—indicating the smallest gap in model accuracy—as representative for each vehicle type. The sensitivity analysis assesses the impact of four critical operational parameters—Speed, Payload of vehicle, Ambient temperature, and Door opening time—on carbon emissions during delivery operations. The bounds for these parameters are established using statistical data and operational insights, ensuring they reflect realistic operational conditions effectively.

- (1) **Speed (v):** Idle times (Speed = 0) are excluded to focus solely on active driving periods. For sensitivity analysis, the highest and lowest average speeds observed in the dataset are used as benchmarks. Each momentary speed value is adjusted by dividing it by the original average speed of the trip and then multiplying it by the maximum or minimum average speed. This approach ensures that the recalculated trip average aligns precisely with the designated maximum or minimum average speed, maintaining the integrity of second-to-second variations.
- (2) **Ambient Temperature (T_{amb}) and Door Opening Time (T):** For these parameters, the 5th and 95th percentiles are also used to establish bounds, depicting extreme operational scenarios.
- (3) **Payload of Vehicle (W_j):** The bounds are determined by considering a near-empty vehicle (basket weight only) and a fully loaded vehicle, reflecting the full operational range of payload conditions.

Fig. 9 illustrates these analyses, indicating that Speed (v) significantly impacts emissions, especially for Truck 5 (18 °C), where emissions increase by 10.40 % at the lower bound and decrease by 25.94 % at the upper bound relative to the baseline. This finding

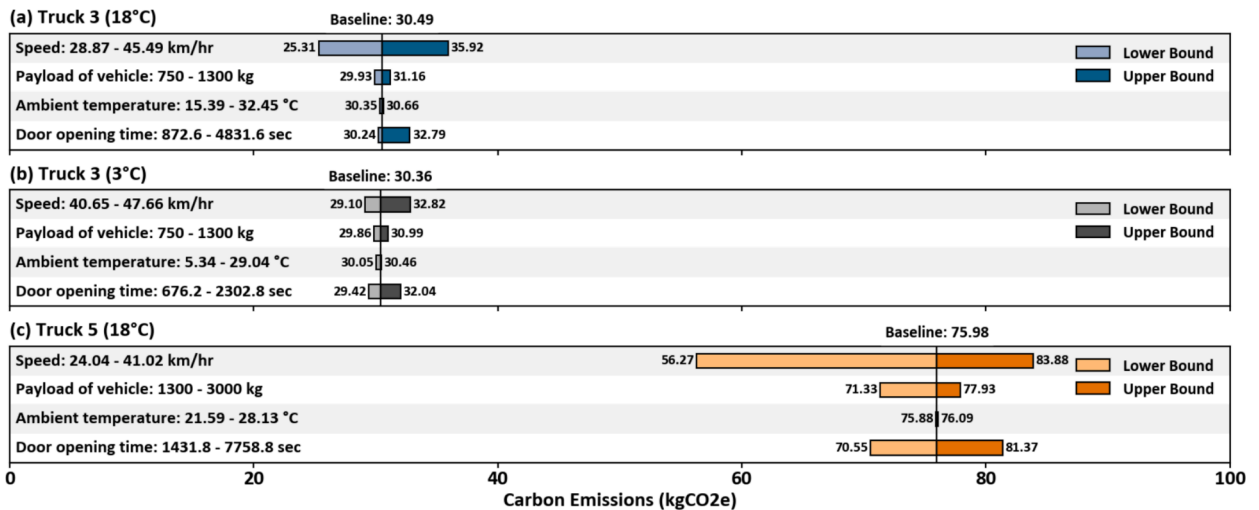


Fig. 9. Sensitivity analysis results for the carbon emissions of delivery segments, focusing on the outcomes of (a) Truck 3 (18 °C) (b) Truck 3 (3 °C), and (c) Truck 5 (18 °C). Note: For each parameter listed, the range is indicated as: Lower bound–Upper bound.

underscores the critical importance of speed optimization in fuel efficiency and emission reduction. Door opening time (T) also shows a notable effect, with emissions adjustments up to +7.57 % and −7.14 %, highlighting the impact of driver operational habits during the loading and unloading phases. Payload (W_i) demonstrates a measurable influence, with emissions increasing by up to 2.57 % when fully loaded and decreasing by 6.12 % when nearly empty. Meanwhile, Ambient Temperature (T_{amb}) had a smaller, yet significant impact, with variations affecting emissions between −1.03 % and +0.32 %.

This sensitivity analysis underscores the dominant influence of driver behaviors, particularly regarding speed regulation and door-opening practices, on carbon emissions. These findings emphasize the need for robust driver training on efficient operational techniques and the integration of technology to monitor and optimize delivery operations. By focusing on these areas, significant reductions in emissions and enhancements in delivery efficiency can be achieved, contributing to the sustainability of logistics operations.

3.5. Evaluation of decarbonization strategies

Our research findings identify potential emission reduction opportunities within the cold chain logistics transportation process. We propose two management strategies aimed at carbon emissions: utilizing shore power for pre-cooling and adopting alternative refrigerants.

(1) Shore power for pre-cooling

Electrification offers a cleaner and more economical alternative to diesel engine idling, substantially reducing carbon emissions from driving, idling, and refrigeration processes. However, the adoption of electric refrigerated vehicles is currently limited by challenges such as insufficient charging networks, a lack of supporting infrastructure, and restricted battery capacity, which collectively impede broader usage (Thornton et al., 2018; Zhang et al., 2023). This study explores a plug-in pre-cooling method that utilizes shore power during the loading phase to pre-cool the TRU, despite the refrigerated trucks not being equipped with batteries. An analysis based on the emission model developed for this research assesses the benefits of employing shore power for pre-cooling. The findings suggest that replacing diesel with shore power for the pre-cooling process can decrease carbon emissions by 53.4 %. Despite the limited current use of fully electric TRUs or refrigerated vehicles, transitioning to shore power for pre-cooling could offer significant long-term energy and economic advantages.

However, implementing shore power for pre-cooling encounters significant obstacles such as high retrofitting costs and the lack of standardized frameworks that restrict widespread adoption (Yin et al., 2020). Variations in government subsidies pose additional financial challenges, but supportive policies could incentivize private enterprises to embrace this technology. Moreover, ensuring the long-term success of this strategy requires a cleaner electricity grid, enhanced by greater reliance on renewable energy sources and advancements in infrastructure, including smart grids and improved energy management practices (Liu et al., 2020; Mac Kinnon et al., 2021).

(2) Adoption of alternative refrigerants

Alternative refrigerants are being recognized as a viable low-carbon pathway for transportation refrigeration, with regulatory bodies increasingly aiming to reduce the environmental impact of refrigerants on ozone depletion and GHG emissions (Wu, Liu, et al., 2022). Commonly used synthetic refrigerants like R404A, R134A, and R410A in vapor compression refrigeration systems are hydrofluorocarbons (HFCs), which are slated for phase-out under EU regulations due to their high GWP (Schulz & Kourkoulas, 2014). There's a growing shift towards natural refrigerants such as hydrocarbons, ammonia, water, air, and carbon dioxide. For instance, R717 (ammonia) and R744 (CO₂), along with mid- to low-GWP refrigerants like R1234yf, are gaining traction (Cardoso et al., 2017). Taking Truck3 (18 °C) delivery as an example, this study compares the impacts of three refrigerants—R410A, R134A, and R744 (CO₂)—against the traditionally used R404A, with GWPs of 2100, 1340, and 1, respectively. The findings, depicted in Fig. 10, indicate that using R410A and R134A could reduce carbon emissions by about 46.46 % and 65.83 %, respectively, during the refrigerated transport process compared to R404A. Switching to R744 (CO₂) could nearly eliminate emissions, reducing them by 99.97 %.

While refrigerant leakage contributes only a minor portion of overall carbon emissions during cold chain transport, the broader environmental impact of refrigerants spans their entire lifecycle—from production to disposal. Over their lifecycle, refrigerant emissions could account for as much as 20 % of the total carbon footprint of cold chain transportation (Bin et al., 2022). Therefore, adopting low-GWP and natural refrigerants is critical to progressing toward a sustainable cold chain system. However, challenges remain, such as flammability, toxicity risks, high-pressure requirements for CO₂ systems, and significant initial costs (Tejani et al., 2022)—particularly in developing regions where access to affordable refrigeration technology is limited (Dong et al., 2021). Overcoming these challenges requires advancements in safety technologies and financial mechanisms like subsidies and carbon credits to facilitate adoption (Sanguri et al., 2021). Although these refrigerants offer substantial environmental benefits, their practical feasibility and the associated costs require careful consideration.

4. Conclusion

This study significantly advances green logistics by innovatively addressing critical gaps in the identification of emission hotspots across cold chain logistics operations. Our approach goes beyond traditional analysis by developing a bottom-up carbon emission model that leverages real-time IoT data from 128 long-haul deliveries. This model analyzes two types of vehicles (3.32-ton and 5.75-ton trucks) under two product temperature categories (18 °C and 3 °C). The analysis spans multiple phases of logistics operations including transport, unloading, and loading, with a specific focus on emissions generated from vehicle driving, idling, refrigeration activities, and refrigerant leakage. Our model validation results underscore its accuracy, revealing an average prediction error of

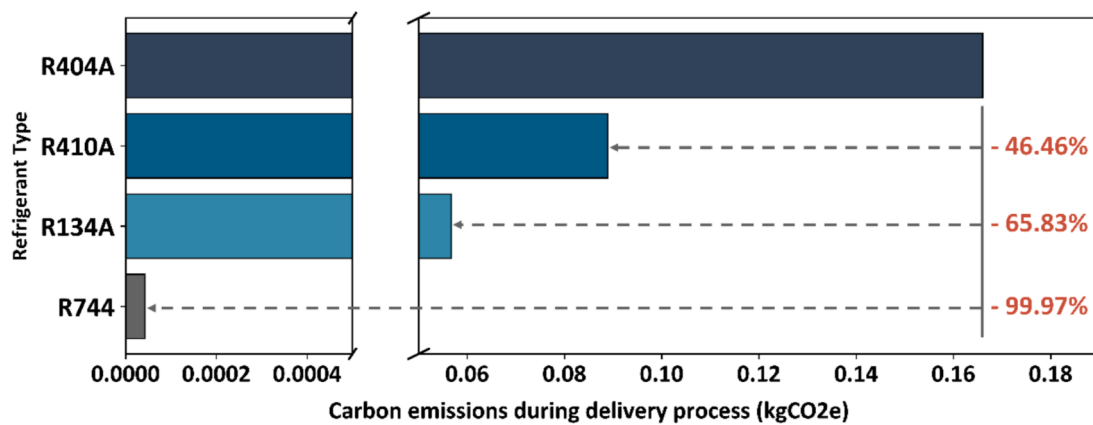


Fig. 10. Comparative analysis of carbon emissions during the refrigerated transport process with different refrigerants.

merely -1.84% compared to actual fuel consumption data.

Our findings reveal several critical emission hotspots and lead to the proposal of multiple carbon reduction strategies. These include the gradual introduction of new energy-efficient refrigerated trucks and the use of shore power for pre-cooling, which offers a cleaner and more economical alternative to diesel engine idling. Despite the lack of batteries in refrigerated trucks, shore power can be utilized during the loading phase to pre-cool the TRU, offering not only reduced carbon emissions but also cost savings and environmental benefits over diesel. Further, we suggest adopting low-GWP refrigerants and emphasize the importance of route optimization to minimize carbon footprints effectively. Enhancing the carbon awareness and training of logistics personnel is also crucial to ensure alignment with green logistics standards. Notably, driving emissions have been identified as the primary source of carbon emissions, contributing significantly ($62.8\% - 66.9\%$) to fuel consumption. However, idling also adds a considerable amount to emissions, especially given its substantial proportion ($60.0\% - 74.0\%$) of the total delivery time.

Refrigeration emissions, influenced significantly by ambient temperature changes, can see transmission emissions fluctuate, particularly near target temperatures, necessitating adaptive refrigeration management strategies. Infiltration emissions, linked to the duration of door openings and influenced by logistics practices, highlight the need for improved operational protocols. Regarding refrigerant leakage, our study promotes the gradual replacement of high-GWP refrigerants with eco-friendly alternatives. Comparative analysis within this research shows that using alternatives like R410A and R134A significantly lowers emissions, while R744 (CO₂) offers the potential to nearly eliminate them. Despite the minor role ($0.9\% - 1.5\%$) of pre-cooling emissions in total outputs, their cumulative environmental impact over the long term should not be underestimated. Transitioning to shore power for pre-cooling not only reduces emissions by over half (53.4%) but also surpasses diesel in terms of economic and environmental returns.

In conclusion, our findings underscore the benefits of employing shore power as a bridging technology towards the adoption of fully electric refrigerated trucks. This research emphasizes crucial improvements needed within the logistics sector, particularly the adoption of innovative refrigerants and the optimization of delivery routes to reduce idling times and energy consumption, which collectively help to minimize the carbon footprint of these operations.

While this study provides valuable insights into reducing carbon emissions within cold chain logistics, there are several opportunities for further research and exploration. This journey involves not only technological advancements but also a blend of regulatory reforms, economic incentives, and significant behavioral changes among stakeholders. Future studies might assess the scalability and economic viability of electric refrigerated trucks across different geographic contexts and explore their integration into existing logistics frameworks (Link & Plötz, 2022). Furthermore, our model focuses on energy consumption during the use phase, excluding the manufacturing and disposal stages of trucks, TRUs, and refrigerants. Future research could assess the full life cycle impacts and operational efficiencies of alternative refrigerants, providing deeper insights into their practicality (Baral et al., 2013). Additionally, engaging more deeply with logistics personnel to understand and improve current operational practices could reveal substantial challenges in adopting these green technologies. Issues such as driver behavior and infrastructure limitations require targeted solutions that can only be developed through detailed, ground-level insights (Filina-Dawidowicz & Csiszár, 2022). Collectively, these strategies not only promote sustainable development within the logistics sector but also contribute to global efforts to achieve net-zero emissions, laying a robust foundation for continued research and practical implementation of sustainable logistics practices.

CRediT authorship contribution statement

Hung-Jui Lin: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Pei-Ci Chen:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Hsuan-Po Lin:** Writing – review & editing, Validation, Resources, Investigation, Data curation, Conceptualization. **I-Yun Lisa Hsieh:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

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Data availability

Data will be made available on request.

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